SkillNavora

PDF 11

Capstone Projects and Portfolio Workbook

Eight capstone briefs, the portfolio checklist and fill-in templates from project charter to README.

Part of the AI for Data Engineers (Data Science & AI) programme.

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Edition: November 2026. Official links point to the provider's own site. The sandbox used to write this could not open them, so check each link before you share it.

Capstone Projects and Portfolio Workbook

Pick a brief, then fill in the templates as you work; they become your write-up.

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C4 Eight Capstone Project Briefs and Portfolio Checklist

Item	Answer
Project 1: Business analytics dashboard	Excel cleaning, SQL analysis, Power BI dashboard, KPI reporting and recommendations. Deliverables: dataset, cleaning log, SQL file, PBIX, one-page summary.
Project 2: Customer churn prediction	Python, pandas, statistics, a classification model, evaluation with precision, recall and ROC-AUC, and a business interpretation with a retention plan.
Project 3: End-to-end data engineering pipeline	Extract from an API, load raw files, transform with dbt or SQL into bronze, silver and gold layers, orchestrate with Airflow or a scheduler, and run data-quality checks. Deliverables: repository, architecture diagram, run instructions, quality report.
Project 4: AI-powered data analyst	Upload a CSV or Excel, ask questions in natural language, generate and validate SQL or Python, create charts and a business summary. Validation: compare generated results with a manual check.
Project 5: Enterprise RAG knowledge assistant	Ingest documents, chunk, embed, store in a vector index, retrieve, answer with citations, evaluate faithfulness and recall on a labelled question set.
Project 6: AI agent for business operations	Design the agent, add tool calls to two business APIs, orchestrate steps, add human approval for risky actions, and monitor each run.
Project 7: MCP-powered business assistant	An MCP server with read tools for a database and one business app, scoped permissions, logs, and an AI client that uses it.
Project 8: Industry-specific capstone (choose one)	Real estate lead and customer analytics; education analytics and an AI learning assistant; financial document processing and analytics; customer service intelligence; e-commerce customer intelligence. Each needs: business problem, users, data sources, requirements, architecture, data flow diagram, implementation steps, tests, metrics, limitations, future work, GitHub repository, demo video and documentation.
Portfolio checklist (all projects)	GitHub repository with README, requirements file, tests, sample data (or a public dataset link), screenshots, a 3-minute demo video, a one-page business summary and a limitations section.

C12 Capstone Workbook Templates (PDF 11)

Item	Answer
1. Project charter	Title · Business question (one sentence) · Who will use the answer · Decision it supports · Success measure (number and target) · Data sources and owners · Scope in / out · Risks and assumptions · Timeline (weeks) · Ethics and privacy check.

2. Data dictionary	For each table: name, grain (one row = ?), source, refresh. For each column: name, type, meaning, allowed values or range, % missing, example value, owner.
3. Data quality log	Check · Rule · Rows failing · Action taken · Date. Examples: duplicate order IDs, negative prices, missing dates, out-of-range ages.
4. Analysis log	Date · Question · Method (query, chart, model) · Result · What it means · Next step. Keep it as you work; it becomes your write-up.
5. Model card (ML projects)	Purpose · Training data and period · Features · Model and settings · Metrics on test data (with baseline) · Fairness checks by group · Known limits · How to monitor.
6. Results summary (one page)	Headline answer in one sentence · 3 key findings with numbers · Recommendation and expected impact · Caveats · Next steps · Link to dashboard or notebook.
7. Presentation outline (7 slides)	1 Question and why it matters · 2 Data and approach · 3–5 Findings (one per slide, chart + takeaway title) · 6 Recommendation and impact · 7 Limits and next steps.
8. Portfolio README	Project title and one-line result · Problem · Data (with source and licence) · Tools · Method · Results with a chart · How to run it · What I would do next · Contact.